CONTROL AND COORDINATION

NCERT · CBSE Class 10 (Science) — Consolidated Question-Answer Set
with Solved Worksheet (MCQs + Assertion-Reasoning)

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About this document. The questions received had several typing errors; they have been corrected (e.g., “reaaly neuron” → relay neuron; “hypothrodium” → hypothyroidism; “corpus collusim” → corpus callosum; “hydroptrposiim” → hydrotropism; “tropicc/nastic” → tropic/nastic) and renumbered Q1–Q16 (the original list skipped Q14). Answers follow the NCERT textbook with supporting standard references. The uploaded worksheet (Grade X, dated 7.08.2026) is solved in Part B, with a quick answer key at the end.

PART A — CORRECTED QUESTIONS WITH ANSWERS (Q1–Q16)

Q1. Differentiate between sensory, motor and relay neurons. Why is a neuron called the structural and functional unit of the nervous system?

Three kinds of neurons work together to conduct messages:

Feature	Sensory neuron (Afferent)	Motor neuron (Efferent)	Relay neuron (Interneuron / Connector)
Direction of impulse	From receptors / sense organs → central nervous system (spinal cord & brain)	From central nervous system → muscles / glands (effectors)	Within the CNS — links sensory neurons to motor neurons
Location	Cell bodies lie in ganglia (e.g., dorsal root ganglion) outside the CNS	Cell bodies lie inside the CNS (spinal cord / brain); long axons run out to effectors	Entirely inside the brain and spinal cord
Function	Carry information about stimuli from receptors to the CNS	Carry instructions from the CNS to muscles and glands to produce the response	Connect sensory and motor neurons; interpret and integrate impulses

A neuron is the structural and functional unit of the nervous system because: structurally, the entire nervous system — brain, spinal cord and nerves — is built up of neurons (each with a cyton/cell body, dendrites and an axon); functionally, every nervous activity — receiving a stimulus (dendrites), conducting the impulse (axon) and transmitting it to the next neuron or to a muscle/gland (synaptic terminal) — is performed only by neurons. No other cell can generate and conduct nerve impulses.

Q2. Why can’t electrical signals directly reach every cell of the body?

- A nerve impulse travels only along nerve fibres (neurons), so an electrical signal reaches only those cells connected to nervous tissue (such as muscles and glands). Most cells of the body have no direct nerve connection.
- Even between two neurons, the impulse cannot jump electrically across the synapse — it must first be converted into a chemical signal (neurotransmitter).
- This is why the body uses a second communication system: hormones (chemical messengers) carried by the blood, which can reach each and every cell of the body, although slowly. Multicellular organisms therefore need both the fast, targeted electrical system (nerves) and the slow, whole-body chemical system (hormones).

Q3. At which site does the neuron convert electrical impulses into chemical signals?

At the synapse — specifically at the axonal (synaptic) terminal of the neuron. When the electrical impulse arrives at the axon terminal, it triggers the release of neurotransmitter chemicals (e.g., acetylcholine) into the synaptic cleft (gap). These chemicals carry the message across the gap to the next neuron — so the signal changes from electrical (inside the neuron) to chemical (at the synapse).

Q4. Which part of the neuron receives information from the axon of another neuron?

The dendrites (dendritic tips). The neurotransmitter released by the axon terminal of the previous neuron binds to specific receptors on the dendrites of the next neuron, and a fresh electrical impulse is generated there.

Q5. Define reflex action.

A reflex action is a sudden, rapid, automatic and involuntary response of the body to a stimulus, produced without the conscious thought of the brain. It is mediated by the spinal cord (or brainstem) through a fixed pathway called the reflex arc. Examples: withdrawing the hand on touching a hot object, blinking when an object approaches the eye, the knee-jerk reflex, sneezing, coughing.

Q6. Write the importance of reflex action.

1. Instant protection: it quickly removes the body from danger (hand withdrawal from heat, closing of eyelids) and prevents serious injury or burns.
2. Maximum speed: because the message does not travel up to the brain and back, the response occurs in a fraction of a second — this has great survival value.
3. Reduces the burden on the brain: the brain stays free for complex, conscious activities while routine emergency responses are handled by the spinal cord.
4. Automatic and reliable: reflex responses work even when one is asleep, distracted or under shock, so the body is always protected.

Q7. “The brain is indirectly involved in reflex action.” Justify the statement.

In a reflex action the reflex arc lies in the spinal cord:

Receptor (skin) → Sensory neuron → Relay neuron (spinal cord) → Motor neuron → Effector (muscle)

The spinal cord itself connects the incoming sensory impulse to the outgoing motor command, so the decision and the response are completed at the spinal-cord level — the brain is not directly involved.

However, sensory impulses are also carried up to the brain through the ascending tracts of the spinal cord. So the brain is informed of what has happened and the person becomes aware of it (e.g., feels the pain) after the reflex has already taken place. Thus the brain’s role is only indirect — it is informed, but it does not control the reflex response.

Q8. Explain why reflex actions are considered more efficient than quick (conscious) responses.

1. Shorter pathway: the impulse travels only from the receptor to the spinal cord and back to the effector; a conscious response would send it all the way to the brain and back, taking much longer.
2. Fewer synapses: each synapse causes a small delay; the reflex arc has fewer junctions, so transmission is faster.
3. No thinking time: no time is lost in perceiving, deciding and voluntarily ordering the movement.

Because of all three, a reflex is completed in a fraction of a second — exactly what is needed to escape injury — while even the quickest conscious response would be slower and might come too late.

Q9. How is the spinal cord protected?

1. It runs inside the neural canal formed by the hollow arches of the vertebrae — the bony vertebral column (backbone) encloses and shields it along its whole length.
2. It is wrapped in three protective membranes called meninges — dura mater, arachnoid and pia mater.
3. The space between the meninges is filled with cerebrospinal fluid (CSF), which acts as a cushion/shock-absorber against jerks and mechanical injuries.
4. It is continuous with the brain, and its upper end is additionally covered by the skull.

Q10. What is corpus callosum? What are its functions?

Definition: The corpus callosum is a thick, C-shaped band of nerve fibres (about 200 million axons, called commissural fibres) lying deep in the brain, which connects the left and right cerebral hemispheres.

Functions:

1. It acts as a bridge that lets the two hemispheres communicate with each other.
2. It transfers and integrates motor, sensory and cognitive information between the two halves of the cerebrum.
3. Since the left hemisphere controls the right side of the body and the right hemisphere controls the left side, the corpus callosum lets the two halves share information so the brain works as one coordinated unit (e.g., coordinated use of both hands, combining visual and language information).

Q11. What is hypothyroidism? How can it be managed? What dietary changes help this condition?

Hypothyroidism is the condition caused by under-secretion of thyroxine by an underactive thyroid gland. Symptoms arise from a slowed metabolism: weight gain, fatigue, feeling cold, dry skin and hair, constipation, slow heartbeat and puffiness. In adults the severe form is myxoedema; in infants it causes cretinism; iodine deficiency can cause simple goitre (enlarged thyroid).

Management:

1. Hormone-replacement therapy — a daily tablet of levothyroxine (synthetic thyroxine, T4), taken exactly as prescribed; the dose is adjusted by the doctor on the basis of regular TSH/T4 blood tests.
2. Treat the cause — e.g., if the cause is iodine deficiency, use iodised salt.
3. Take the tablet correctly — on an empty stomach, 30–60 minutes before breakfast, with plain water; keep iron/calcium supplements and soy foods at least 4 hours away from the dose.

Dietary changes that help:

1. Ensure adequate but not excess iodine — iodised salt, fish, dairy products, eggs.
2. Include selenium and zinc — Brazil nuts, fish, eggs, sunflower seeds (selenium helps convert T4 into the active T3 form).
3. Eat a balanced, nutrient-dense diet — vegetables, fruits, whole grains and lean protein — with regular meals and exercise, since a low metabolic rate makes weight control difficult.
4. Limit excess raw goitrogens (soy products, cabbage, cauliflower, broccoli) especially if iodine intake is low; cooking these vegetables reduces their goitrogenic effect.
5. Keep high-fibre foods, coffee, and calcium/iron pills away from the medicine time (they reduce absorption), and avoid crash diets.

Q12. The thymus gland is large in children and shrinks at puberty. Why?

The thymus is the ‘training school’ of the immune system. Immature lymphocytes made in the bone marrow travel to the thymus, where they mature into T-lymphocytes (T-cells) and learn to distinguish the body’s own cells from foreign invaders. A large, active thymus is therefore essential

during childhood, when the immune system is being built for the first time.

By puberty a sufficient population of mature, 'trained' T-cells has already been produced and stationed in the lymph nodes and spleen, so the thymus has completed its main job. The sex hormones (androgens and oestrogens) released at puberty then trigger gradual thymic involution — the gland shrinks and its lymphoid tissue is progressively replaced by fat. It continues to shrink with age, though it keeps producing some T-cells throughout life.

Q13. A hormone X is secreted into the blood when a person is under a scary situation. Identify the hormone and the gland.

Hormone X is adrenaline (epinephrine), secreted by the adrenal gland — specifically its inner part, the adrenal medulla — located on top of each kidney. Adrenaline is the "emergency" or "fight-or-flight" hormone: it increases the heart rate and breathing rate, raises blood pressure, and diverts more blood and glucose to the skeletal muscles so the body can face the danger or run away from it.

Q14. How are the timing and the amount of release of insulin regulated in our blood?

Insulin release is regulated by a feedback (negative-feedback) mechanism, with the beta cells of the islets of Langerhans in the pancreas acting as the glucose sensor:

1. After a meal, blood glucose rises → the beta cells sense it and release more insulin.
2. Insulin makes the liver and muscle cells take up glucose and store it as glycogen, so blood glucose falls back to normal.
3. As glucose returns to normal, the beta cells get the signal and reduce / stop insulin release.
4. If glucose drops too low, the alpha cells release glucagon, which raises it again.

Thus when insulin is released and how much is released are set automatically by the prevailing blood-glucose level — the feedback loop keeps glucose within the normal range.

Q15. Differentiate between tropic and nastic movements.

Basis	Tropic movements	Nastic movements
Direction	Directional — the movement depends on the direction of the stimulus	Non-directional — the movement does not depend on the direction of the stimulus
Nature	Growth movements — caused by cell division and cell elongation in a particular zone	Usually caused by a change in turgor (water content) of cells — not by growth
Speed / reversibility	Slow, permanent (irreversible)	Quick, generally reversible
Examples	Phototropism (shoot bends to light), geotropism, hydrotropism, chemotropism, thigmotropism (tendrils)	Drooping of touch-me-not (Mimosa pudica) leaves on touch; opening/closing of flowers with light

Q16. Design an experiment to demonstrate hydrotropism.

Aim: To show that roots grow towards water (positive hydrotropism).

Materials: a wide tub/trough filled with dry sand, soaked pea (or gram) seeds / a seedling, one small porous clay pot, water.

Procedure:

1. Fill a wide tub with dry sand and sow a few germinating seeds (or plant a seedling) near the centre.
2. Fill the porous clay pot with water and bury it in the sand on one side of the seedling.
3. Leave the set-up undisturbed for a few days, adding water only to the porous pot (so moisture comes from one side only).

4. Control: in another tub, water the sand uniformly on all sides so there is no one-sided water source.

Observation: In the experimental set-up, the roots bend and grow towards the porous pot (the source of water) instead of growing straight down. In the control, the roots grow straight downwards.

Conclusion: Roots grow towards water, demonstrating positive hydrotropism — here the root responds to the moisture gradient even more strongly than to gravity.

PART B — SOLVED WORKSHEET

Worksheet: Control and Coordination — Grade X (Section A, dated 7.08.2026)

Section 1 (Q1): 15 MCQs · Section 2 (Q2): 10 Assertion-Reasoning items. Each answer below is followed by a one-line reason.

B1. Multiple-Choice Questions — Answer Key

Q1	**b**	**Q2**	**b**	**Q3**	**b**	**Q4**	**c**	**Q5**	**b**
Q6	**d**	**Q7**	**c**	**Q8**	**b**	**Q9**	**b**	**Q10**	**b**
Q11	**b**	**Q12**	**b**	**Q13**	**b**	**Q14**	**b**	**Q15**	**c**

Q1 — Answer (b). A nerve impulse travels as an electrical signal from the dendritic end towards the axonal end; at the synaptic terminal it releases neurotransmitter chemicals (electrical → chemical).

Q2 — Answer (b). The conversion of an electrical signal into a chemical signal occurs at the axonal (synaptic) end.

Q3 — Answer (b). The cerebellum maintains posture and balance and gives precision to voluntary actions such as walking in a straight line.

Q4 — Answer (c). Impulse path: Dendrite → Cell body → Axon → Axonal end.

Q5 — Answer (b). Blood pressure, salivation and vomiting are involuntary actions controlled by the medulla in the hindbrain.

Q6 — Answer (d). Abscisic acid inhibits growth and causes wilting of leaves under environmental stress.

Q7 — Answer (c). Both are tropic: the sunflower bends towards sunlight (phototropism) and the tendril coils around a support (thigmotropism).

Q8 — Answer (b). Stem: positively phototropic and negatively geotropic; root: negatively phototropic and positively geotropic.

Q9 — Answer (b). Iodine deficiency → thyroid gland affected → goitre (thyroxine synthesis needs iodine).

Q10 — Answer (b). Growth hormone is secreted by the pituitary gland; its hypo-secretion causes dwarfism.

Q11 — Answer (b). Precise coordination of voluntary hand movements is the job of the cerebellum.

Q12 — Answer (b). On the shaded side auxin concentration increases, stimulating greater cell elongation, so the shoot bends towards light.

Q13 — Answer (b). High glucose → pancreatic cells secrete insulin → liver/muscle cells take up glucose and store it as glycogen (feedback control).

Q14 — Answer (b). Reflex arc: Receptor (skin) → Sensory neuron → Spinal cord (relay neuron) → Motor neuron → Effector (leg muscle).

Q15 — Answer (c). Cytokinin is abundant in rapidly dividing tissues (seeds, fruits) and promotes cell division.

B2. Assertion-Reasoning — Answer Key

Options: (a) Both true, R explains A · (b) Both true, R does not explain A · (c) A true, R false · (d) A false, R true

A&R; 1	a	A&R; 2	a	A&R; 3	a	A&R; 4	c	A&R; 5	a
A&R; 6	a	A&R; 7	a	A&R; 8	a	A&R; 9	c	A&R; 10	c

1 — Answer (a). Mimosa's touch response (seismonasty) is independent of growth, and R correctly explains it — cells change shape through turgor (water) changes, causing swelling/shrinking.

2 — Answer (a). Neurotransmitters are stored and released only from axon terminals, and their receptors are present only on the dendrites of the next neuron — hence transmission across a chemical synapse is one-way (unidirectional).

3 — Answer (a). Iodine is needed to synthesise thyroxine, which regulates carbohydrate, protein and fat metabolism — exactly why iodised salt is recommended.

4 — Answer (c). A is true. R is false: the part of the tendril in contact with the support grows slower than the part away from it (by elongation, not rapid division) — that is why it coils around the support.

5 — Answer (a). Reflexes are mediated at the spinal-cord level, and R explains the advantage — an extremely rapid survival response before the brain can process the input.

6 — Answer (a). Adrenaline is the emergency/fight-or-flight hormone because it raises heart rate, breathing rate and blood pressure and diverts blood to skeletal muscles — R explains A.

7 — Answer (a). The medulla oblongata houses the vital centres for respiration, heartbeat and blood pressure — damage to it is fatal; R explains A.

8 — Answer (a). Without a nervous system or muscles, plants still sense and respond directionally because they use electrical-chemical signals and change cell water content or elongation rates — R explains A.

9 — Answer (c). A is true (insulin is injected subcutaneously to lower blood glucose). R is false: insulin is secreted by the beta cells of the pancreas (islets of Langerhans) and it converts glucose into glycogen.

10 — Answer (c). A is true. R is false: auxin accumulates on the shaded side, and the shaded-side cells grow longer — that is why the shoot bends towards light.

QUICK ANSWER KEY (Worksheet)

MCQs (Q1–Q15): 1-b · 2-b · 3-b · 4-c · 5-b · 6-d · 7-c · 8-b · 9-b · 10-b · 11-b · 12-b · 13-b · 14-b · 15-c

Assertion-Reasoning (1–10): 1-a · 2-a · 3-a · 4-c · 5-a · 6-a · 7-a · 8-a · 9-c · 10-c

Reference: NCERT Class 10 Science — Control and Coordination (reflex arc, neuron, tropic/nastic movements, plant and animal hormones); standard endocrinology references for hypothyroidism management (levothyroxine, iodine/selenium nutrition), thymic involution, corpus callosum and the insulin feedback loop.